CompTIA Core 2 220-1102

Instructor: Andrew Ramdayal

CompTIA A+ CORE 1 (220-1102) Exam

- **90-minute** time limit
- Maximum of **90 Questions**
  - Multiple choice
    - Pick one or many answers.
  - Drag & Drop
    - Match objects to a diagram.
  - Performance-based (Simulators)
    - These are hands-on troubleshooting scenarios where you'll have to perform a series of steps/commands
- **700 (78%)** out of a scale of 100-900

Domain	% of Exam
1.0 Operating Systems	31%
2.0 Security	25%
3.0 Software Troubleshooting	22%
4.0 Operational Procedures	22%

Operating Systems

1.1

Operating Systems (OS)

- An operating system(OS) is required to get any functionality out of a computer system.
- There are many types of OSes, each with their own features and functions but they all provide these basics.
- Common Functions
  - Interface to interact with the system
  - Drivers to communicate to the hardware
  - Applications to provide additional functionality
  - File Management features to copy, move, and delete files
  - Network Connectivity to connect to local resources and the internet
  - System Security to prevent access from unauthorized users

Operating System Types

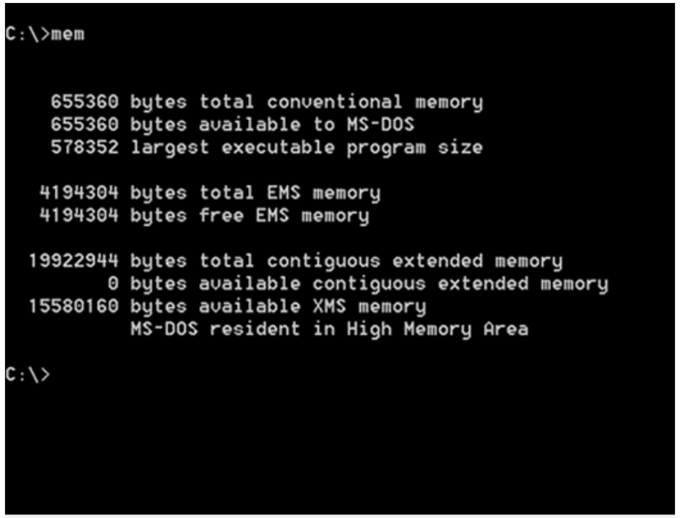
- Closed Source operating systems are only available from a single organization.
- Open-Source operating systems can be distributed by many different organizations and the code can be freely modified.

Operating System Types

Operating System	License	Platform
Microsoft Windows	Closed Source	Desktop or laptop computer
Apple macOS	Closed Source	Desktop or laptop computer
Linux	Open Source	Desktop or laptop computer
Google Chrome OS	Closed Source	Desktop or laptop computer
Google Android	Open Source	Mobile device
Apple iOS/iPadOS	Closed Source	Mobile device

User Interfaces

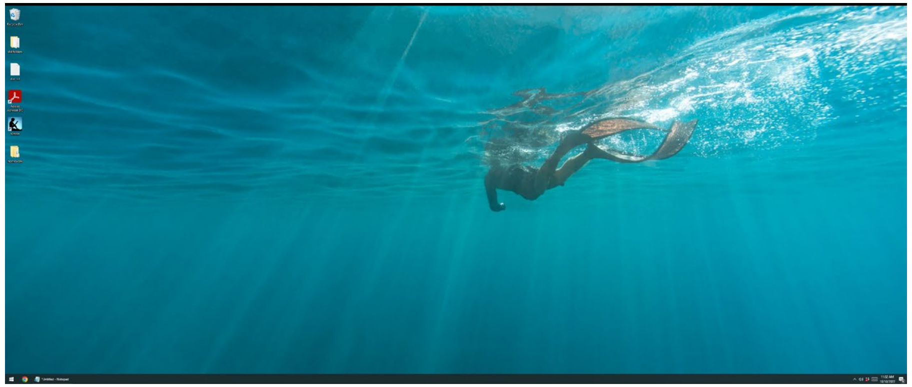
- Command Line Interface (CLI): This kind of OS is controlled by typing commands into a prompt. The most commonly known Command Line OS is DOS and Cisco's IOS.



MS DOS 6.2 CLI

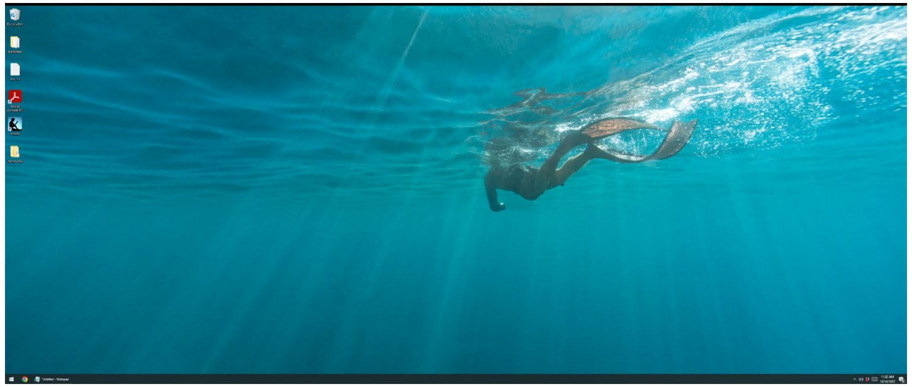
User Interfaces

- Graphical User Interface (GUI):
  - This OS uses graphics.
  - provides the icons we click on and the mouse pointer that lets us click on them.
  - Every version of Windows has a GUI that we use to interact with it.



User Interfaces

- Graphical User Interface (GUI):
  - This OS uses graphics.
  - provides the icons we click on and the mouse pointer that lets us click on them.
  - Every version of Windows has a GUI that we use to interact with it.



Windows 10

- Released in 2015 and will be supported till 2035

Windows 10 Hardware requirements

Component	32-bit/x86	64-bit/x64
CPU	1 GHz	1 GHz
RAM	1 GB	2 GB
Storage	16 GB	32 GB
Graphics	DirectX 9 or higher	DirectX 9 or higher

Windows 10 Editions

- It is essential to pick an edition of Windows that is appropriate for the user not just the one with the most features.
- This minimizes the waste or system resources and money.

Windows 10 Editions

Feature	Windows 10 Home
Target group	Consumer users
Network support	Workgroup
NTFS Encryption	❌ No
Remote Desktop	Client only
Hyper-V	❌ No
BitLocker	❌ No
Group Policy	❌ No
Physical CPU Limit	1
RAM Limit	128 GB

Windows 10 Editions

Feature	Windows 10 Pro
Target group	Corporate users
Network support	Domain and Workgroup
NTFS Encryption	✔ Yes
Remote Desktop	Client and server
Hyper-V	✔ Yes
BitLocker	✔ Yes
Group Policy	✔ Yes
Physical CPU Limit	2
RAM Limit	2 TB

Windows 10 Editions

Feature	Windows 10 Pro for Workstations / Enterprise
Target group	Corporate users
Network support	Domain and Workgroup
NTFS Encryption	✔ Yes
Remote Desktop	Client and server
Hyper-V	✔ Yes
BitLocker	✔ Yes
Group Policy	✔ Yes
Physical CPU Limit	4
RAM Limit	6 TB

Upgrade Installation (In Place Upgrade)

- Replace Windows but keep all your data and compatible applications in place.
- Requirements
  - A previous bootable version of Windows already installed
  - Installation media on removable media or stored locally.
- Upgrade considerations
  - Backup files and user preferences
  - Application and driver support/backward compatibility
  - Hardware compatibility

Upgrade Installation (In Place Upgrade)

- In-place upgrade to Windows 10
  - Windows 7
  - Windows 8.1 (Upgrade 8.0 to 8.1)
- In-place upgrade to Windows 11
  - Windows 10

Upgrade Installation (In Place Upgrade)

- Upgrade considerations
  - Backup files and user preferences
  - Application and driver support/backward compatibility
  - Hardware compatibility
  - Downgrade from a Pro 7 to a home 10 will result in losing certain setting.

1.2

Navigating Commands

- **help** : is used to list commands
- **dir** : list files and folders
- **/?:** Give help on the command
- **cd(chdir)** : is used to move from one folder to another
- **md(mkdir**) : is used to make new folders
- **rd(rmdir)** : is used to delete empty folders
- **del** : is used to delete files
- **tree** : list files and folder within the current folder and all sub folders
- Drive navigation inputs: **C:** or **D:**
- **Winver :** Shows what version of windows you are on
- **cls** : Clear screen

Copy Commands

- **copy** : is used to copy files from one folder to another
- **xcopy** : it can copy folders, subfolders, and all the files with them
- **robocopy** : more advanced copy task than xcopy

Copy Commands

- **copy** : is used to copy files from one folder to another
- **xcopy** : it can copy folders, subfolders, and all the files with them
- **robocopy** : more advanced copy task than xcopy

Networking Commands

- **ping**: Uses ICMP to return the status of a unicast.
  - **ping –n**: change the number of pings sent.
  - **ping –l**: change the size of the ping packets.
  - **ping –t**: pings continuously
  - **ping –4**: force an IPv4 ping
  - **ping –6**: force an IPv6 ping

Networking Commands

- **ipconfig**: Displays interface configurations.
  - **ipconfig /all**: displays more detailed information.
  - **ipconfig /renew**: request configurations from a DHCP server.
  - **ipconfig /release**: removes configurations obtained through DHCP.
  - **ipconfig /displaydns**: displays the local DNS cache.
  - **ipconfig /flushdns**: clears the local DNS cache.

Networking Commands

- **net use**: Used to connect to a network share
  - **net use x: \\servername\sharename**
- **netstat**: Displays active network connections
  - **netstat –a**: displays all connections, including listening ports
- **tracert**: Uses ICMP to return a hop count.
- **net user**: Used to manage user accounts

Networking Commands

- **pathping:** Performs a ping and a traceroute at the same time
- **nslookup**: Identifies the current DNS server and displays IP Addresses for a provided name.
- **hostname:** displays a computer's hostname

Storage Commands

- **Chkdsk :** scans a disk in hopes of recovering corrupted files
  - **chkdsk /f** Fixes errors on the disk
  - **chkdsk /r** Fixes errors and Locates bad sectors
- **Format :** erases a disk file applying a file system
  - **format /fs** Specifies the type of the file system (FAT, FAT32, exFAT, NTFS)
    - **format d: /fs:ntfs** will format the D drive as ntfs
  - **format /q** Performs a quick format
    - **format d: /fs:fat32 /q** will quick format the D drive as fat32

Storage Commands

- **Convert :** changes FAT/FAT32 filesystem to NTFS without erasing files
  - **convert /fs** Specifies that the volume will be converted to NTFS
    - Cannot convert NTFS to any other file system
- **Diskpart :** is a command line partition management tool

Administration Commands

- **sfc** is the system file checker used to repair system files.
  - **sfc /SCANNOW** Scans integrity of all protected system files and repairs files.
  - **sfc /VERIFYONLY** Scans system files but does not repair them.
- **shutdown** will send a signal to turn off the sytem
  - **shutdown /p** Turn off the local computer with no time-out or warning.
  - **shutdown /r** Full shutdown and restart the computer.

Administration Commands

- **Gpupdate** : Updates the latest group policy setting
- **Gpresult** : shows what group policy is applied to the computer

1.3

Task Manager

- **Task Manager**
  - **Processes**
    - Displays all running processes including background processes.
    - Non-responsive processes can be closed here.
  - **Performance**
    - Displays performance graphs
  - **Users**
    - Displays currently logged-in users
    - It is possible to log out users in this tab
  - **Startup**
    - Disable or enable auto-starting applications
- **`taskmgr`** can be used to launch the task manager via a run box

Microsoft Management console (mmc)

- **Microsoft Management Console**
  - Create a custom toolbox of useful utilities referred to as “Snap-ins”.
  - **Snap-ins** are other consoles that are available elsewhere like the Device Manager or Disk Management.
  - **mmc** command can be used to launch the Microsoft Management Console.

Useful Snap-in's

- **Event Viewer (eventvwr.msc)**
  - display logs of timestamped events which can be used to assist with troubleshooting.
  - **Windows Logs**
    - **System:** list operating system events
    - **Security:** list security events
    - **Application:** list application events
  - **Icons**
    - **Red** = Error
    - **Yellow** = Warning
    - **White** = Informational

Useful Snap-in's

- **Disk Management (diskmgmt.msc)**
  - Manage Disk partitions
- **Task Scheduler (taskschd.msc)**
  - Create and schedule tasks to run
- **Device Manager (devmgmt.msc)**
  - Check, update and install device drivers
- **Certificate Manager (certmgr.msc)**
  - Check and manage certificates installed on a computer

Useful Snap-in's

- **Local Users and Groups (lusrmgr.msc)**
  - Create, change and delete users on local computer
- **Performance Monitor (perfmon.msc)**
  - Monitor computer performance
- **Group Policy Editor (gpedit.msc)**
  - Edit local group policy

System Information

- **System Information**
  - View detailed information on system hardware and software.
  - **msinfo32** command can be used to launch the System Information utility.
  - **Sections**
    - **Hardware Resources**
      - Identify hardware conflicts and addresses
    - **Components**
      - Identify driver details and hardware capabilities
    - **Software Environment**
      - Identify software details.

Additional Tools

- **System Configuration (msconfig.exe)**
  - **General**
    - Change startup type between Normal, Selective, or Diagnostic types.
  - **Boot**
    - Change multiboot boot order
  - **Services**
    - Enable or disable services
  - **Startup**
    - Links to the Startup tab in the task manager
  - **Tools**
    - Collection of useful tools

Additional Tools

- **Disk Cleanup (cleanmgr.exe)**
  - Files in the Recycle Bin
  - Temporary Internet files
  - Downloaded program files
  - Temporary files
- **Disk Defragment(dfrgui.exe)**
  - Optimize and Defragment Drives (Windows 10)
  - **Defragmenting (HDD)**
    - Bits on a hard disk drive are rearranged so files can be loaded faster.
      - Defragging a drive too frequently can decrease its lifespan.
  - **Trimming (SSD)**
    - Makes sure that the NAND memory chips on an SSD are worn evenly to maximize the lifespan of the drive.

Additional Tools

- **Registry Editor (regedit.exe)**
  - a database that stores all the settings and configurations for Windows and it’s applications.
  - The **`regedit`** command can be used to launch the **Registry Editor**
- **Registry Keys**
  - **HKEY_CLASSES_ROOT:** Stores **file association** information.
  - **HKEY_USERS:** Stores settings that apply to **all users**.
  - **HKEY_CURRENT_USER:** Stores settings for the **individual users**.
  - **HKEY_LOCAL_MACHINE:** Stores settings for **all devices** that have been installed or removed from the system.
  - **HKEY_CURRENT_CONFIG:** Stores settings for individual devices when **multiple** of the same type of device have been installed.

1.4

Windows 10 Control Panel utility

- **Internet Options**
  - Configure default internet browser options
- **Devices and Printers**
  - Add, remove and administrator printers, scanners, cameras and other devices.
- **Programs and Features**
  - Reinstall, uninstall programs and windows features.
- **Network and Sharing Center**
  - Check and administer NIC
- **System**
  - Check computer specification, rename computer, join domain or workgroup

Windows 10 Control Panel utility

- **Windows Defender Firewall**
  - Check and change firewall setting. Can open ports.
- **Mail**
  - Add, remove, or repair mailboxes. Mostly used by Microsoft Outlook.
- **Sound**
  - Use to setup speaker or mic's on a computer.
- **User Accounts**
  - Use to change, add, or remove local user accounts.

Windows 10 Control Panel utility

- **Device Manager**
  - Check if devices are functioning correctly. Update or rollback drivers
- **Indexing Options**
  - Check what is being index on a system
- **Administrative Tools**
  - Set of commonly used utilities to manage the system
- **Ease of Access**
  - Make the system easier to use for persons with disabilities
- **File Explorer Options**
  - Show hidden files
  - Hide extensions
  - General options
  - View options

Windows 10 Control Panel utility

- **Power Options**
  - Hibernate
  - Power plans
  - Sleep/suspend
  - Standby
  - Choose what closing the lid does
  - Turn on fast startup
  - Universal Serial Bus (USB) selective suspend

1.5

Windows settings

- **Time and Language**
  - Configure time and date, and language used on the computer.
- **Update and Security**
  - Set when updates will be applied to the computer.
- **Personalization**
  - Personalization of the system for to the user likening such as background
- **Apps**
  - Uninstall applications, change windows defaults, and enable or disable windows features.
- **Privacy**
  - Set what can be tracked on the system.

Windows settings

- **System**
  - Allows you to change display information, sound, and notification setting.
- **Devices**
  - Manage Bluetooth, printers, and a mouse.
- **Network and Internet**
  - Manage and connect new NIC
- **Gaming**
  - Connect Xbox gaming accounts
- **Accounts**
  - Create and link new accounts to the system

1.6

Workgroup vs. Domain

- **Workgroup**
  - Decentralized setup used in SOHO
  - Uses local user accounts
  - No central server for computer or user management
  - Simple to setup with no additional server software needed

Workgroup vs. Domain

- **Domain**
  - Centralized setup used in small-large businesses
  - User accounts are manage on a central server called domain controllers
  - Computer configuration and security setting are set on a central server
  - Need to setup a server (Windows Server), more expensive

Workgroup vs. Domain

- **Shared resources**
  - Folder or devices shared on a network
- **Printers**
  - Printers shared on a network
- **File servers**
  - Shares a folder for other computers to access
- **Mapped drives**
  - Allows a shared folder on another computer to act as a drive on a system.

Firewall Settings

- **Firewalls**
  - Block all incoming traffic
  - Allows all outgoing traffic
  - Configure and manage with rules
  - Will need to make an exception to allow certain traffic such as ftp through the firewall

Client network configuration

- **Internet Protocol (IP) addressing scheme**
  - Assign by the network administrator
  - E.g 192.168.10.10
- **Domain Name System (DNS) settings**
  - Assign by the network administrator
  - E.g 1.1.1.1 (cloudflare DNS Server)
- **Subnet mask**
  - Assign by the network administrator
  - E.g 255.255.255.0
- **Gateway**
  - Assign by the network administrator
  - E.g 192.168.10.1
- **Static vs. dynamic**
  - Assign by the network administrator
  - Static is manually typed in by a technician vs. Dynamic is assign by the DHCP Server.
  - If no DHCP is available when selecting dynamic the cpmptuer will assign APIPA address of 169.254.x.x

Network connections

- **Establish network connections**
  - **Virtual private network (VPN)**
    - Allows you to access a remote network over the internet
  - **Wireless**
    - Connects to a local network using a wireless connection
  - **Wired**
    - Connects to a local network using an ethernet cable
  - **Wireless wide area network (WWAN)**
    - Internet access using a wireless connections. Done by using an adapter from a mobile cellular network using technologies such as 4G or 5G.

Network connections

- **Proxy settings**
  - A server used to control and monitor internet access
  - Configuration is given by the administrator
- **Public network vs. private network**
  - When connecting to a network you will select either setting.
  - Public will offer more protection while private will allow shares and discovery of the computer.
- **File Explorer navigation – network paths**
  - Allows you to map a network drive from file explore
- **Metered connections and limitations**
  - Limits the amount of data that can be sent and receive on an interface

1.7

Installing Applications

- **32-Bit vs. 64-Bit Requirements**
  - 32 bit processors can handle large amounts of RAM vs. 32-bit.
  - 32-bit can use only about 4GB of RAM
  - 64-Bit can use 16 exabytes of RAM.
  - 64-Bit will require a 64-bit processor and operation system
    - Check Windows to check if you are running a 64-bit OS.
  - 64-Bit operating system can run 32-bit application
  - 32-bit operating systems cannot run a 64-bit application

Installing Applications

- **Requirements when installing Applications**
  - Dedicated graphics card vs. integrated
    - Some application will require more higher end graphics to run such as games
  - Video random-access memory (VRAM) requirements
    - Memory build into the graphics cards.

Installing Applications

- **Requirements when installing Applications**
  - RAM requirements
    - Check the RAM requirements before purchasing the application

Installing Applications

- **Requirements when installing Applications**
  - Central processing unit (CPU) requirements
    - Check the CPU requirements before purchasing the application

Installing Applications

- **Requirements when installing Applications**
  - External hardware tokens
    - USB stick used to access the application.

Installing Applications

- **Distribution methods**
  - Physical media vs. downloadable
    - Physical media uses DVD or USB's.
    - Downloadable are EXE files downloaded from a site
  - ISO mountable
    - An image of a disk.
    - Single file that stores all the necessary files for the application

Installing Applications

- **Considerations for applications**
  - Impact to device
  - Impact to network
  - Impact to operation
  - Impact to business

1.8

Workstations Operating Systems

- Windows 10
  - World's most used desktop operating systems.
  - Used in both businesses and homes
- Linux
  - Uses by my power users and servers
  - Open Source mostly can be downloaded for free
- macOS
  - Mostly used by home or small business users
- Chrome OS
  - A Linux-based operating system that uses Chrome as its main interface

Cell phone/tablet OSs

- iPadOS
  - Used on Apple's Ipad devices
- iOS
  - Uses on Apple Iphone's
- Android
  - Uses on other manufacture mobile devices such as Samsung, Sony, or Goolge

Windows File Systems

File System	Advantages	Disadvantages
FAT32 File Allocation Table 32	• Most compatible file system	• Partitions are limited to 32GB • File are limited to 4GB • No security features

NTFS
New Technology File System
	• Supports partitions bigger than 32GB
	• Supports files bigger than 4GB
	• Disk quotas
	• File and folder compression
	• File system security
	    • File and folder permission
	    • EFS (Encrypting file system)
		• Only officially compatible with Windows

exFAT Extensible File Allocation Table	• More compatible than NTFS • Supports partitions bigger than 32GB • Supports files bigger than 4GB	• No security features

Non-Windows File Systems

- **macOS File Systems**
  - **HFS+** (Hierarchical File System Plus)
  - **APFS** (Apple File System)
  - macOS does support read and write access to FAT32 and exFAT partitions but only support read-only access to NTFS partitions.
- **Linux File Systems**
  - **ext3** (Third Extended File System)
  - **ext4** (Fourth Extended File System)
  - Linux can read and write to NTFS, FAT32, exFAT, and HFS+.
- **Optical Disc File Systems**
  - **CDFS** (Compact Disc File System)
  - **UDF** (Universal Disc Format)

File System Compatibility

File System	Windows Support	macOS Support	Linux Support
FAT32	✔Yes	✔Yes	✔Yes
NTFS	✔Yes	Read Only	✔Yes
exFAT	✔Yes	✔Yes	✔Yes
HFS+	✘No	✔Yes	Read Only
APFS	✘No	✔Yes	✘No
ext3	✘No	✘No	✔Yes
ext4	✘No	✘No	✔Yes
CDFS	✔Yes	✔Yes	✔Yes
UDF	✔Yes	✔Yes	✔Yes

Vendor life-cycle limitations

- All operating systems have an End-of-life (EOL)
  - When the manufacture stop supporting the operating system.
  - Windows 7 EOL was 1/14/202
  - Windows 10 EOL will be 10/14/2025
- Once it reaches it’s EOL their will no updates to the OS.

Concerns

- Compatibility concerns between Oss
  - Application are developed to run on a specific OS.
  - Some application has different version for Windows or Mac and some don't

1.9

Installing an OS

- Check the following before attempting to Install Windows:
  - CPU
  - RAM
  - Storage Requirement

Installing an OS

- Boot methods
  - USB
    - OS files stored on a USB Stick
  - Optical media
    - OS files stored on a DVD or Blue Ray
  - Network
    - Files are stored on a network server and download to the compute when installing. Used on large deployments.
    - WDS (Windows Deployment Service) running on a Windows Server
    - The target computer must support network booting often noted as PXE (Preboot Execution Environment).

Installing an OS

- Boot methods
  - Solid-state/flash drives
    - OS files stored on a USB Stick
  - Internet-based
    - Files are downloaded from the internet when installing.
    - Mostly used on linux.
  - External/hot-swappable drive
    - OS files stored on an external Hard drive.
  - Internal hard drive (partition)
    - OS files stored on an internal drive

Types of installations

- Clean Installation
  - This is the most common way to install Windows onto a single PC. Requirements
  - An empty hard drive
  - Bootable installation media (DVD/USB)

Types of installations

- Upgrade Installation (In Place Upgrade)
  - This is the easiest option if you just want to replace Windows but keep all your data and compatible applications in place.
  - Requirements
    - A previous bootable version of Windows already installed
    - Installation media on removable media or stored locally.
  - Upgrade considerations
    - Backup files and user preferences
    - Application and driver support/backward compatibility
    - Hardware compatibility

Types of installations

- Network Install/Deployment (OS Deployment)
  - This is the fastest way to install Windows on many computers since you're doing over the network.
  - Requirements
    - WDS (Windows Deployment Service) running on a Windows Server
    - The target computer must support network booting often noted as PXE (Preboot Execution Environment).

Types of installations

- Cloning / Imaging (Ghosting)
- Duplicates the entire software installation of a system. This includes the operating system, drivers, applications, and configurations.
- This can be done by directly connecting a hard drive or over the network.
- Before you clone the drive, you must run “SysPrep” to remove security IDs that are generated for activation purposes.
- Recovery Partition Installation
- Pre-Built systems sold with an operating system already installed will either include a recovery partition with the operating system, drivers, and other bundled software.
- Repair Installation

Types of installations

- Cloning / Imaging (Ghosting)
  - Duplicates the entire software installation of a system. This includes the operating system, drivers, applications, and configurations.
  - This can be done by directly connecting a hard drive or over the network.
  - Before you clone the drive, you must run “SysPrep” to remove security IDs that are generated for activation purposes.

Post-Installation

- Install OS Updates to assure the system has the latest features and security updates.
- Upgrade Drivers to manufactures latest drivers is required.
  - Microsoft has greatly improved relationships with hardware manufactures so they can deliver updated drivers.
- Restoring User Data files that are required for their work from their older system or a backup.
- Feature updates
  - Product life cycle

Partition Table Formats

- Logical segments of a physical hard drive.
- Created for data separation.
- **MBR (Master Boot Record)**
  - This is the **first sector** of a MBR partitioned drive and contains code that informs the system about installed OS.
  - Allows for **4 primary** partitions.
  - Limited to **2.2 TB** partitions
- **GPT (GUID Partition Table)**
  - Theoretically allows for **unlimited primary** partitions.
  - Windows is limited to 128 primary partitions by design.
  - NOT limited to 2.2 TB partitions.

Partition

- **Primary Partitions**
  - These partition are used to boot an operating system. If you have multiple operating systems on one disk they each will require their own primary partitions.
- **Extended Partitions**
  - These partition are used to overcome the four primary partition limit.
  - A single extended partition can contain many logical drives, each logical drive appears as a partition but can not be used to store the OS.
- **Hidden partition**
  - Often used by OEMs to store system recovery data (recovery partitions).
- **Swap partition**
  - Used a virtual memory by some operating systems.

Drive format

- **Full Format**
  - Runs an additional step that checks the hard drive for any bad sectors.
- **Quick Format**
  - Drive is not checked for bad sectors

1.10

Installing Software in macOS

- **macOS** includes an **App Store** which includes free and paid applications
  - Application can be downloaded and installed from a vendor's website, but it is NOT enabled by default
    - Must be enabled in system preferences
  - **.pkg** files are compressed files used to install a macOS application
  - **.dmg** files are Apple Disk Image files often used to store compressed software installers
  - **.app** files are installed applications
  - Must have an apple ID to setup and download apps.

MacOS Best practices

- **Backups**
  - As often as data is changing or as must as you are will to lose
- **Antivirus**
  - Should have 3rd party antivirus installed
- **Updates/patches**
  - Install updates as apple releases them

System Preferences

- Displays
  - Configure display setting such as resolution or multiple monitors
- Networks
  - Set network configuration
- Printers
  - Add, manage or remove printers
- Scanners
  - Add, manage or remove Scanners
- Privacy
  - Manage privacy settings
- Accessibility
  - Configure the system for people with disabilities
- Time Machine
  - Backup mechanism of macOS,

Features

- Multiple desktops
  - **Use Mission Control to create additional desktops, called spaces, to organize the windows**
- Mission Control
  - View and manage all open application windows
- Keychain
  - Stores your passwords and account information, and reduces the number of passwords you have to remember and manage.
- Spotlight
  - Finds items on your Mac, like apps, files, and emails

Features

- iCloud
  - Backup and synchronize your photos, files, backups, and more across all your devices
- Gestures
  - Apple trackpad or a Magic Mouse with your Mac, you can use gestures.
  - Click, tap, pinch, and swipe
- Finder
  - Default file manager and graphical user interface shell used on all
- Remote Disc
  - Allows the Mac user access to a CD or DVD disc loaded into a separate computer
- Dock
  - Convenient place to access apps and features that you're likely to use every day

Features

- Disk Utility
  - Can be used to partition and initialize storage devices.
  - It also is used to access First Aid which can repair permissions and recover corrupted files.
- FileVault
  - disk encryption program
- Terminal
  - Unix command line for MacOS
- Force Quit
  - Press these three keys together: Option, Command, and Esc (Escape) or
  - Choose Force Quit from the Apple menu in the corner of your screen.

1.11

Installing Software in Linux

- **Linux** distros include a **Package Manager** to install and update applications.
  - Application can also be installed from other sources like a vendor's website
- Tools
  - Shell/terminal
    - Configure the from a command line
  - Samba
    - Samba is a free software implementation of the CIFS/SMB networking protocols that supports Microsoft Windows Server Domain, Active Directory and Microsoft Windows NT domains.
    - With Samba, Unix-like OSes can interoperate with Windows and provided file and print services to Windows clients.

Linux Commands

- **man:** display help for commands
- **ls:** list files in a directory
- **mv:** move or rename a file
- **cp:** copy a file
- **rm:** remove a file
- **mkdir:** create a new folder
- **df:** display free storage space
- **ps:** displays a list of running processes
- **top:** displays running processes and resource usage
- **find:** search for text in files in a directory
- **grep:** search with regular expression
- **dig:** troubleshoot DNS
- **cat:** displays the contents of a file
- **nano:** text editor
- **pwd:** set or change a password
- **ip:** displays and configure a NIC

Installing Software in Linux via Command Line

- Advanced Packaging Tool (APT) is a command line utility used to install, uninstall, and upgrade applications in Debian-based distributions like Ubuntu
  - apt-get update is used to update the version list of installed applications.
  - apt-get upgrade is used to install the latest version of installed applications.
  - apt-get install is used to install a new application
  - apt-get install chromium-browser
  - apt-get remove is used to uninstall an application.
  - apt-get remove chromium-browser
- The Yellowdog Updater, Modified (YUM) is a free and open-source command-line package-management utility for computers running RedHat-based distributions like Cent-OS

Ownership in Linux and macOS

- Linux and macOS both share origins with another OS known as Unix, because of this they have many similarities when it comes to controlling file access.
  - **chown** is used to change the owner and group of a file.
  - Syntax : chown user:group file
    - **chown** juan:instructors class_presentation
      - This command changes the owner to juan and the group to instructors.
    - **chown** andrew class_presentation
      - This command just changes the owner to andrew.

Ownership in Linux and macOS

- chmod is used to change the permissions of a file or folder
  - Owner/Group/Everyone
  - Owner permissions apply to the original file creator.
  - Group permissions apply to the group of accounts that have been given access.
  - Everyone permissions apply to all accounts… everyone
- Read/Write/Execute are the different permissions that can be granted to a user or a group of users.
  - Read (r) +4 permission allows someone to view the contents of a file.
  - Write (w) +2 permission allows someone to save changes to a file.
  - Execute (x) +1 permission allows someone to execute programs or scripts.
  - To assign full permission you assign read + write + execute or 4 + 2 + 1 = 7
  - To assign just read and write you use 4 + 2 = 6.
  - chmod 764 class_presentation or chmod u+rwx,g+rw,a+r class_presentation
    - Sets owner to have full permissions, the group to have read and write, and everyone to have just read.

Best practices

- Backups
  - As often as data is changing or as must as you are will to lose
- Antivirus
  - Should have 3rd party antivirus installed
- Updates/patches
  - Install updates as apple releases them

Best practices

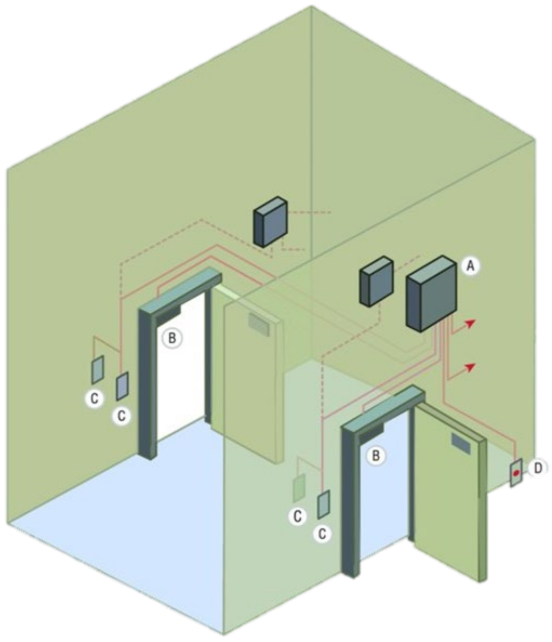
- Backups
  - As often as data is changing or as must as you are will to lose
- Antivirus
  - Should have 3rd party antivirus installed
- Updates/patches
  - Install updates as apple releases them

2.0

2.1

Physical security

- Access control vestibule
  - Controls access so only one person can enter at a time
  - Prevents tailgating and piggybacking



Access Control Vestibule

Physical security

- Badge reader
  - are used to read the data from authentication cards
    - RFID cards are commonly used

Physical security

- Video surveillance and
  - **IP Cameras** have replaced the older analog video surveillance systems
    - **NVR (Network Video Recorder)** is used to aggregate all the IP camera feeds into a single interface
  - **CCTVs** are older analog video surveillance systems

Physical security

- Alarm systems
  - Uses sensors to check if door or windows are open.
  - Checks for motion

Physical security

- Motion sensors
  - Detects physical movements
    - types of motion sensors that are used frequently:
    - **Passive Infrared (PIR)**
    - **Microwave**
    - **Dual Tech/Hybrid**

Physical security

- Door locks
  - Use to lock doors

Physical security

- Equipment locks
  - Construction equipment, trailer, and cargo theft prevention

Physical security

- Guards
  - Human security guards

Physical security

- Bollards
  - A bollard is a sturdy, short, vertical post.
  - Prevent automotive vehicles from colliding or crashing into pedestrians and structures

Physical security

- Fences
  - a barrier, railing, or other upright structure made of any material enclosing an area of ground to control or prevent unauthorized access to the front of the qualified residence.

Physical security for staff

- Key fobs
  - Use to enter doors
- Smart cards
  - Use cards to enter spaces
- Keys
- Biometrics
  - Retina scanner
  - Fingerprint scanner
  - Palmprint scanner
- Lighting
  - Must have adequate lighting to ensure people are visible to cameras and guards
- Magnetometers
  - metal detector is the most-used form of airport security

Physical security for staff

- Key fobs
  - Use to enter doors
- Smart cards
  - Use cards to enter spaces
- Keys
- Biometrics
  - Retina scanner
  - Fingerprint scanner
  - Palmprint scanner
- Lighting
  - Must have adequate lighting to ensure people are visible to cameras and guards
- Magnetometers
  - metal detector is the most-used form of airport security

Logical security

- Principle of least privilege
  - Users should only be given access to the level required for their work
- Access control lists (ACLs)
  - List of rules on a device that defines who can access that device

Logical security

- **Multifactor Authentication (MFA/2FA)** requires users to provide 2 or more types of authentication factors to gain access.

Something you know	• Username & Password, PIN, Answer to a security question
Something you have	• Smart card, Email, Hard token, Soft token, Short message service (SMS), Voice call, Authenticator application
Something you are	• Biometric authentication; fingerprints, handprint, retina scan, palm vain scan
Somewhere you are	• Based on geolocation
Something you do	• Perform a specific action

Logical security

- Hard token
  - Hardware device used to generate a number used to login.

- Soft token
  - Same as a physical token but just an app on a mobile phone

Mobile device management (MDM)

	Who owns the device?	Who controls and maintains the device	Description
BYOD Bring your own device	The user	The user	The user brings their own personally owned device. This provides more user freedom, lower cost to the organization, along with greater risk since the organization does not control, secure, or manage the device.
Corporate Owned	The organization	The organization	The device can only be used for work related to the organization. It can NOT be used for personal use.

Mobile Device Management

- **Application management** features are important to allow enterprise control of applications.
- **Content management** (sometimes called MCM, or mobile content management) ensures secure access and control of organizational files including documents and media on mobile devices.
- **Remote wipe** capabilities are used when a device is lost, stolen, or when the owner is no longer employed by the organization.

Mobile Device Management

- **Geolocation** and geofencing capabilities allow you to use the location of the phone to make decisions about its operation.
- **Screen locks, passwords, and pins** are all part of normal device security models to prevent unauthorized access.
- **Biometrics** are widely available on modern devices, with fingerprints and facial recognition being the most broadly adopted and deployed.
- **Full device encryption (FDE)** remains the best way to ensure that stolen or lost devices don’t result in a data breach.
- **Push notifications** may seem like an odd inclusion here but sending messages to devices can be useful in several scenarios.

Active Directory

- Domain refers to the entire network under the control of the domain controller
  - The network is identified by its domain name
- **Login scripts** are used to automate actions when users log in
- **Group Policies** can be applied to all users and devices from the server
- **Organizational units** are used to group users and devices to simplify management
  - Users are grouped by Role (Sales, HR, Account)
- **Home folders** are private folders users can use to store personal files
- **Roaming profiles** are downloaded to any system the user logins to and then any changes are uploaded back to the server when a user logs out
- **Folder redirection** allows a user's profile data to be accessible when they login to a system without the need to download files. This can speed up the process of login in and out for large profiles

2.2

Protocols and encryption

- WPA (Wi-Fi Protected Access) is more secure than WEP but still vulnerable
  - Users authenticate using an alphanumeric passphrase (PSK) via TKIP(Temporal Key Integrity Protocol)
  - Encrypts with RC4 (Rivest Cipher 4)
  - It takes about 15 minutes to crack WPA

Protocols and encryption

- WPA2 (Wi-Fi Protected Access 2) is more secure than WEP and WPA
  - Users authenticate using an alphanumeric passphrase (PSK) via CCMP(Counter mode Cypher block chaining Message authentication code Protocol)
  - Encrypts with AES (Advance Encryption Standard)

Authentication

- Centralized Authentication Protocols used in businesses to authenticate users to WIFI, VPN, and other network resources:
  - Remote Authentication Dial-In User Service (RADIUS)
  - Terminal Access Controller Access-Control System (TACACS+) (Cisco)
- Kerberos
  - Used on Windows to authenticate users in Active directory
- Multifactor
  - Combine multiple methods to increase security of loggin.

2.3

Malware

- **Viruses**
  - Malware that can self-copy and self-replicate but **requires human interaction** to spread
  - **Virus Types**
    - **Memory resident** viruses, remain in memory while running
    - **Boot sector** viruses, reside in the first sector of storage media, which stores boot data
    - **Macro viruses**, take advantage of automation features in productivity software and spread through files associated with them.
    - **Email viruses**, spread either as attachments or scripts that are part of the email.
- **Worms**
  - Malware that can spread **without human interaction**.
  - Worms can spread from one device in a network to another

Malware

- **Ransomware**
  - A kind of malware that **encrypts a victim's data** and holds the decryption key for ransom
  - An effective backup system that stores data offline.
    - Air gapped data is data that is NOT connected (offline)
- **Cryptominers**
  - Uses a victim's system to **mine for cryptocurrency** without their permissions
  - Doesn’t get aggressive to avoid detection

Malware

- **Trojans**
  - A type of malware that is typically disguised as legitimate software. Software should only be installed from trusted sources.
- **Rootkits**
  - Allows an attacker to execute commands at an **elevated privilege**
  - The best ways to prevent rootkits are normal security practices, including patching, using secure configurations, and ensuring that privilege management is used.
  - Tools like secure boot and techniques that can validate the integrity of live systems and files can help prevent rootkits from being successfully installed or remaining resident.

Malware

- **Spyware**
  - **Obtains information** about an individual, organization, or system and then sends it to a malicious actor.
  - Spyware is most frequently combated using anti-malware tools
  - User awareness training can help prevent the installation of spyware that is included in trojans
- **Keyloggers**
  - Are programs that **capture keystrokes** usually to steal personal data like passwords and financial information
  - Antimalware tools should be able to detect know keylogger malware

Prevent Malware

- **Recovery mode**
  - Microsoft Windows Recovery Environment (Windows RE) is a simplified, scaled-back version of the Windows operating system.
- **OS reinstallation**
  - Reinstalls the OS completely
  - This will remove all malware but you will lose all files and settings

Prevent Malware

- **Use software to detect, clean and prevent malware:**
  - Antivirus
  - Anti-malware
  - Must be kept updated with new signature
- **Software firewalls**
  - Windows Defender Firewalls can prevent worms or virus from entering open ports on a computer.

Prevent Malware

- **User Training**
  - Anti-phishing training
  - User education regarding
    - common threats
  - Can be done to large groups or one-on-one
  - Can use video or live training

2.4

Social Engineering

- Uses social tactics to trick users into giving up information or performing actions they wouldn't usually take.
  - Social engineering attacks can occur in person, over the phone, while browsing the net, or via email.
  - Social engineers take advantage of normal social behaviors and trust

Social Engineering

- **Phishing** is the practice of sending emails to users with the purpose of tricking them into revealing personal information or performing a compromising action.
  - **Phishing** does NOT target a specific group or user which can make it easier to detect
  - **Spear Phishing** targets specific groups of users.
    - More dangerous than standard phishing as the attack can be highly customized
  - **Whaling** targets high-level executives.
    - The individuals being targeted generally have access to very sensitive data
  - **Vishing** is a form of phishing that uses voice.
    - Always verify the identity and contact information of any caller
    - Caller-ID is NOT reliable as it can be spoofed
  - **Smishing** uses SMS (text) messages
    - Includes instant messaging and social messages

Social Engineering

- **In-Person techniques**
  - **Dumpster diving** is when a threat actor searches through trash looking for information. Shredding or burning documents mitigates this threat.
  - **Shoulder Surfing** is looking over someone's shoulder either in person or with a camera in hopes of viewing sensitive information.
  - **Tailgating** is the practice of one person following closely behind another to enter a secure area without showing credentials.

Social Engineering

- **Impersonation**
  - Pretends to be someone else
  - Usually they impersonate tech support personal or company executives
- **Evil Twin**
  - Fraudulent Wi-Fi access point that appears to be legitimate but is set up to eavesdrop on wireless communications.
  - The evil twin is the wireless LAN equivalent of the phishing scam.
  - This type of attack may be used to steal the passwords of unsuspecting users, either by monitoring their connections or by phishing, which involves setting up a fraudulent web site and luring people there.

Threats

- **DoS (Denial of Service)** is an attack that sends a large number of packets in hopes of overwhelming a system so it can no longer provide its service. A DoS is one to one attack.
  - **Ping of Death** when large fragmented ICMP is used to overwhelm a host.

Threats

- **DDoS (Distributed Denial of Service)** is just like a DoS except there are many attackers and one victim. A **traffic spike** is usually the sign that a network is undergoing a DDoS attack.
  - **Botnet:** Network of victim computers under the control of the attacker.
    - This network is usually made of malware victims (Trojans) that are unaware that their systems are part of an attack.
  - **Coordinated Attack:** A command and control server is used to command a Botnet to coordinate the DDoS attack.
  - **Friendly/Unintentional DoS:** Sometimes users may bring down service just by sharing a link on social media. If the link goes viral and the server can't handle the load, it will come down.

Threats

- **Zero-day Attack**
  - A zero-day is a vulnerability being exploited out in the wild but there is no known fix for
- **Spoofing**
  - Spoofing is a technique an attacker uses to hide their identity
  - **ARP Spoofing**
    - Attacker spoofs the IP to MAC mapping usually to perform a man-in-the-middle attack
  - **IP Spoofing**
    - Attackers impersonate a device by IP address
  - **Email Spoofing**
    - Attackers send email messages using email addresses that a target might trust

Threats

- **On-Path Attack (aka Man-in-the-Middle Attack)**
  - Network traffic is intercepted
- **DNS Poisoning**
  - False DNS information
- **ARP Spoofing**
  - Attacker spoofs the IP to MAC mapping usually to perform a man-in-the-middle attack

Threats

- **Brute-Force attacks** attempt to defeat a password using automated random guessing
  - Long and complex passwords will increase the amount of time it will take the attacker to guess the password
  - Can be prevented by limiting the number of consecutive attempts
  - Can always succeed given enough time
- **Dictionary attacks** use a list of known passwords
  - Not using common words and phrases will make this attack more difficult
  - Avoid reusing passwords to limit the effectiveness of this attack

Threats

- **Insider Threat**
  - An insider threat is a trusted person (employee, contractor, partner) who commits a malicious act
- **Cross-Site Scripting (XSS)**
  - An attacker injects malicious code into a website through an insecure form
- **SQL Injection Attacks**
  - An attacker compromises a SQL database usually through cross-site scripting
  - **Structured Query Language (SQL)** is used to create, store, and retrieve information from a database

Vulnerabilities

- Non-compliant systems
- Unpatched systems
- Unprotected systems
  - missing antivirus/missing firewall
- EOL OSs
- Bring your own device (BYOD)

2.5

Windows OS Security Setting

- Defender Antivirus
  - Activate/deactivate
  - Updated definitions
- Firewall
  - Activate/deactivate
  - Port security
  - Application security
- Users and groups
  - Local vs. Microsoft account
  - Standard account
  - Administrator
  - Guest user
  - Power user

Windows OS Security Setting

- Login OS options
  - Username and password
  - Personal identification number
  - (PIN)
  - Fingerprint
  - Facial recognition
  - Single sign-on (SSO)

NTFS vs. Share Permissions

- Permissions can be set on a folder using both NTFS and sharing option
- The most restrictive will apply
- Inheritance
  - Files and folders will inherit it's permission from a parent folder

Opening Apps

- Run as administrator vs. standard user
  - Certain applications will require an admin login
  - User Account Control (UAC)

Encryption Setting

- BitLocker
  - Full volume encryption feature included with Microsoft Windows
  - Protect data by providing encryption for entire volumes.
  - BitLocker To Go
    - Drive Encryption on removable data drives
- Encrypting File System (EFS)
  - Provides filesystem-level encryption.
  - Enables files to be transparently encrypted to protect confidential data from attackers with physical access to the computer

2.6

Workstation Security

- Data-at-rest encryption
  - Use bitlocker or EFS to encrypt data stored on the computer
- Password best practices
  - Complexity requirements
    - Length – 8-10 Min
    - Character types – Mix of all character on keyboard
  - Expiration requirements
    - Should expire every 60-90 days
  - Should set a BIOS or UEFI Password
    - This must be entered before windows boot.

Workstation Security

- End-user best practices
  - Use screensaver locks
  - Log off when not in use
  - Secure/protect critical hardware
  - (e.g., laptops)
  - Secure personally identifiable information (PII) and passwords
- Disable AutoRun and AutoPlay

Workstation Security

- Account management
  - Restrict user permissions
    - Give Min permission to do the job
    - No one should have an admin account
  - Restrict login times
    - Should only be able to login during work hours
  - Disable guest account
  - Use failed attempts lockout
  - Use timeout/screen lock
  - Change default administrator's user account/password

2.7

Mobile Device Security

- Screen locks
  - Facial recognition
  - PIN codes
  - Fingerprint
  - Pattern
  - Swipe
- Remote wipes
  - If lost, can remotely wipe the device
- Locator applications
  - Able to find the device if lost

Mobile Device Security

- OS updates
  - Keep the device updated with the latest updates from the manufacture
- Device encryption
  - Full disk encryption
  - Most newer phones has this on by default
- Remote backup applications
  - Ability to remotely backup data on the device
- Failed login attempts restrictions
  - If the passcode is entered to many times wrong the device will lock or be wipe
- Antivirus/anti-malware
  - Generally 3rd party software to prevent or clean malware
- Firewalls
  - Helps to protect worms or virus from entering the device

Mobile Device Security

- Policies and procedures
  - BYOD vs. corporate owned
  - Profile security requirements
- Internet of Things (IoT)
  - All device connected to the internet.
  - Change default passwords
  - Keep updated

2.8

Data Destruction

- Physical destruction
  - Drilling
    - Using a drill to break the platter in the drive
  - Shredding
    - Uses a device to physically shred the drives into small pieces
  - Degaussing
    - Using a large magnetic to remove data from the disk
  - Incinerating
    - Melts the drive

Data Destruction

- Recycling or repurposing best practices
  - Erasing/wiping
  - Low-level formatting
  - Standard formatting
- Outsourcing concepts
  - Third-party vendor
  - Certification of destruction/recycling

2.9

SOHO Network Security

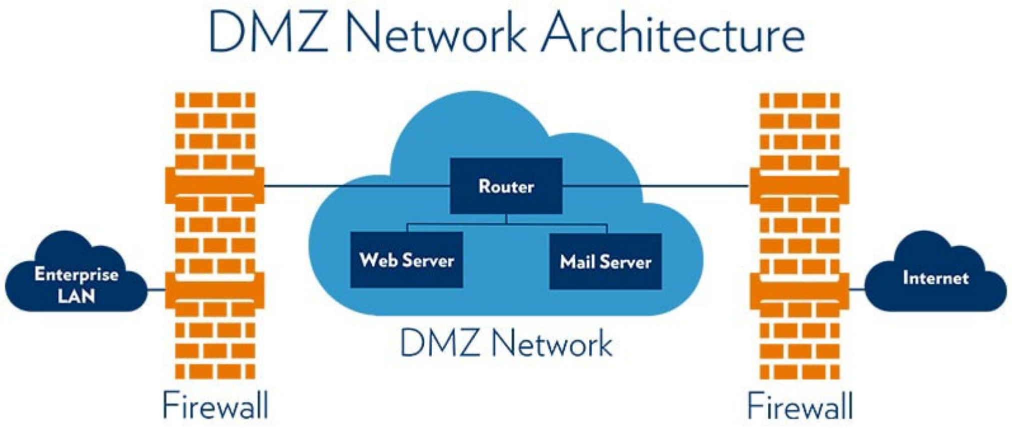
- Home router settings
  - Change default passwords
  - IP filtering and Content Filtering
    - Filter unwanted content from IP Address or sites
  - Firmware updates
    - Should be kept updated since newer firmware will include security updates
  - Physical placement/secure locations
    - Should be stored in a secured location to ensure no authorized physical access to the device

SOHO Network Security

- Home router settings
  - Dynamic Host Configuration Protocol (DHCP) reservations
    - To ensure a certain device such as a printer always receive a set IP Address
  - Static wide-area network (WAN) IP
    - If your ISP gives you a static IP address it will have to be configured on the router
  - Universal Plug and Play (UPnP)
    - Enables apps and devices to automatically open and close ports to connect with the LAN network
  - Screened subnet
    - A demilitarized zone where companies store publicly accessible servers such as a web server

SOHO Network Security

- Screened subnet
  - A demilitarized zone where companies store publicly accessible servers such as a web server



SOHO Network Security

- Wireless specific
  - Changing the service set identifier (SSID)
  - Disabling SSID broadcast
  - Encryption settings
  - Disabling guest access
  - Changing channels

SOHO Network Security

- Firewall settings
  - Disabling unused ports
  - Port forwarding/mapping
    - Enable remote access to applications or server from outside the network

2.10

Browsers Security

- Browser download/installation
  - Trusted sources
    - Hashing
  - Untrusted sources
- Extensions and plug-ins
  - Trusted sources
  - Untrusted sources
- Password managers
- Secure connections/sites –valid certificates

Browsers Security

- Settings
  - Pop-up blocker
  - Clearing browsing data
  - Clearing cache
  - Private-browsing mode
  - Sign-in/browser data synchronization
  - Ad blockers

3.0

3.1

Common troubleshooting steps

- Reboot
- Restart services
- Uninstall/reinstall/update applications
- Add resources
- Verify requirements
- System file check
- Repair Windows
- Restore
- Reimage
- Roll back updates
- Rebuild Windows profiles

Tools to Fix Windows

- Windows Repair
  - Windows Recovery environment
- Windows Reset
  - Reinstalls Windows but allows you to keep your files
- System Restore
  - Allows the system to restore back to a date
- Safe Mode
  - Boot’s the system with minimum drivers and software to check the operating system

Common troubleshooting steps

- Reboot
  - Easiest and faster way to fix simple issue
  - May work for frozen OS or application
- Restart services
  - Will refresh the entire service
  - May work for a service that is not functioning or has failed
  - Can be done from the **services.msc**

Common troubleshooting steps

- Uninstall/reinstall/update applications
  - Best uninstall the application completely then reinstall
  - This will fix most issues with an application
  - Application data files might not be lost
  - Might lost application setting
  - Can be done from the **control panel** or **Apps** from **setting**

Common troubleshooting steps

- Verify requirements
  - Before installing any OS or application ensure the system meets the minimum requirements.
  - Many applications and OS has a minimum amount of RAM, Disk Space and CPU requirements

Common troubleshooting steps

- System file check
  - Allows users to scan for and restore corrupted Windows system files
  - Use the SFC command
    - sfc /scannow
      - Scans all system files and replaces corrupted or incorrect files.

Common troubleshooting steps

- Reimage
  - Uses an imaging software to create an image
  - When issues occur, the system will be reimage

Common troubleshooting steps

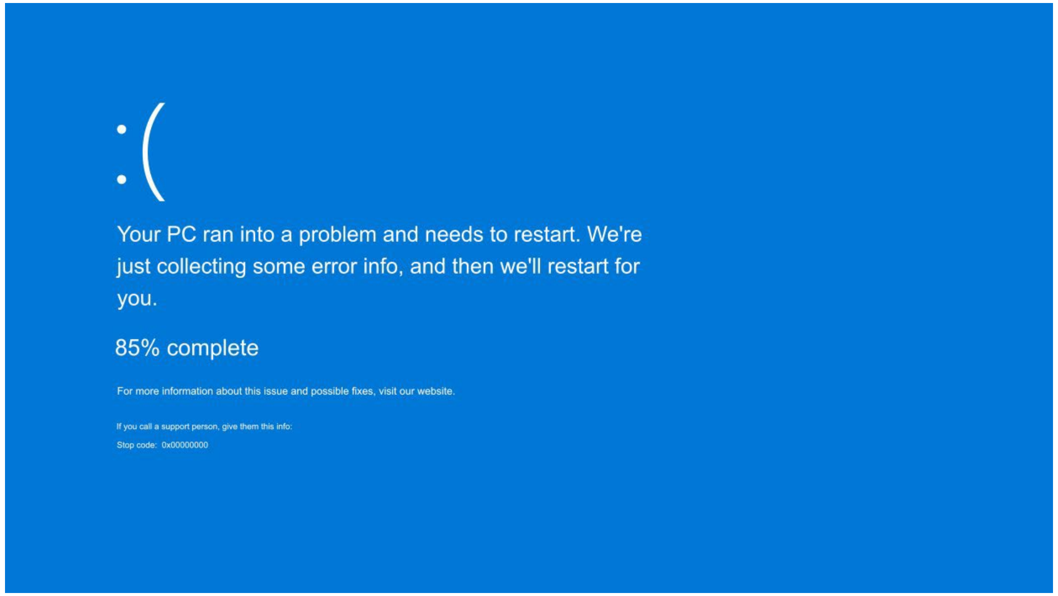
- Roll back updates
  - If updates causes errors, then you can remove the update if needed

Common troubleshooting steps

- Rebuild Windows profiles
  - Builds a new windows profile for the users
  - This can resolve any issues with the user application or configuration

Common Windows OS Problems

- Blue screen of death (BSOD)
  - Windows Issues
    - Corrupt OS files or drivers
  - Hardware failure such as RAM
  - Application error



Common Windows OS Problems

- Sluggish performance
  - Computer running slow
  - Check you have minimum amount of resources
  - Check task manager for how much resources is being used by applications
  - Reinstall application or add more RAM, SSD, or faster processor
  - Reinstall OS as last option

Common Windows OS Problems

- Frequent shutdowns
  - Windows Issues
    - Corrupt OS files or drivers
  - Hardware failure
    - RAM failure
    - Cooling issues (Fans not working)

Common Windows OS Problems

- Services not starting
  - Issues with the service itself
  - Best to reinstall the application that installed the service
  - Use system restore and restore to point when it was working

Common Windows OS Problems

- Applications crashing
  - Check if any application setting has recently change
  - Backup the application data and reinstall the application
  - Use system restore and restore to point when it was working

Common Windows OS Problems

- Low memory warnings
  - Not enough RAM to run the application
  - Check task manager to see what application is using how much memory
  - Add more RAM, that way you use less page memory

Common Windows OS Problems

- USB controller resource warnings
  - May happen if too many devices are connect to a USB Bus
    - Plug some devices into USB 2.0 ports
    - Reinstall the USB host controller driver

Common Windows OS Problems

- System instability
  - System crashing, slow, or error
  - Could be caused by corruption in Windows OS files
    - Use SFC /scannow

Common Windows OS Problems

- No OS found
  - Boot loader for Windows is corrupt or cannot be found
  - Check if your booting to correct drive
  - Check if there is something plug into a boot drive that is set in the bios
  - Use startup repair
  - Use the command below to rebuild the Windows Boot Configuration database
    - Bootrec /rebuildbcd

Common Windows OS Problems

- Slow profile load
  - When using logs in, it can take very long for the desktop to load
  - Check what application are starting up
  - If it is a roaming profile
    - Check to ensure you have enough bandwidth to load the profile

Common Windows OS Problems

- Time drift
  - Date and time maybe off
  - Set the computer to update the time and date automatically

3.2

Computer (PC) Security Issues

- Unable to access the network
  - Can be malware is slowing down the system
- False alerts regarding antivirus protection
  - Says no antivirus is installed or expired.
  - Says your computer is infected and needs to download a false antivirus

Computer (PC) Security Issues

- Altered system or personal files
  - Missing/renamed files
  - Typical for malware to rename files or altered them
- Unwanted notifications within the OS
  - Might get notification the OS is not functioning correctly
- OS update failures
  - Computer cannot be updated
- Random/frequent pop-ups
  - Pop-ups from websites or from malware on your computer

Browser-related symptoms

- Certificate warnings
  - Certificate from site is expired or not trusted
  - Don’t visit site that gives this error
- Redirection
  - Going to bad website it will redirect you to another site that has malware

3.3

Procedures for Malware Removal

1. Investigate and verify malware symptoms
   - Pop-up
   - Message asking for money
   - Not booting
   - Slow
   - Applications not working

Procedures for Malware Removal

2. Quarantine infected systems

- Remove the system off the network
- Disconnect the NIC or disable the wifi card.

Procedures for Malware Removal

2. Quarantine infected systems
    - Remove the system off the network
    - Disconnect the NIC or disable the wifi card.
3. Disable System Restore in Windows
4. Remediate infected systems
    a. Update anti-malware software
    b. Scanning and removal techniques (e.g., safe mode, preinstallation environment)
5. Schedule scans and run updates
6. Enable System Restore and create a restore point in Windows
7. Educate the end user

3.4

Mobile OS and Application Issues

- Application fails to launch
  - Uninstall and reinstall App
- Application fails to close/crashes
  - Uninstall and reinstall App
- Application fails to update
  - Uninstall and reinstall App
- Slow to respond
  - Update device
  - Remove apps that could be consuming the resources
- OS fails to update
  - Check OS setting
  - Factory reset

Mobile OS and Application Issues

- Battery life issues
  - Applications running in the background
  - Stop them from running
  - Running the phone in high performance mode
- Randomly reboots
  - Corruptions of the OS or getting too hot

Mobile OS and Application Issues

- Connectivity issues (ensure it is enable)
  - Bluetooth
    - Disconnect and reconnect device
    - Ensure within range
  - WiFi
    - Disconnect and reconnect device
    - Ensure within range
  - Near-field communication (NFC)
    - Ensure within range
  - AirDrop
    - Ensure within range
- Screen does not autorotate
  - Ensure phone is set to rotate
  - Accelerometer is faulty

3.5

Mobile OS and Application Security Issues

- Security concerns
  - Android package (APK) source
  - Developer mode
  - Root access/jailbreak
  - Bootleg/malicious application
  - Application spoofing

Mobile OS and Application Security Issues

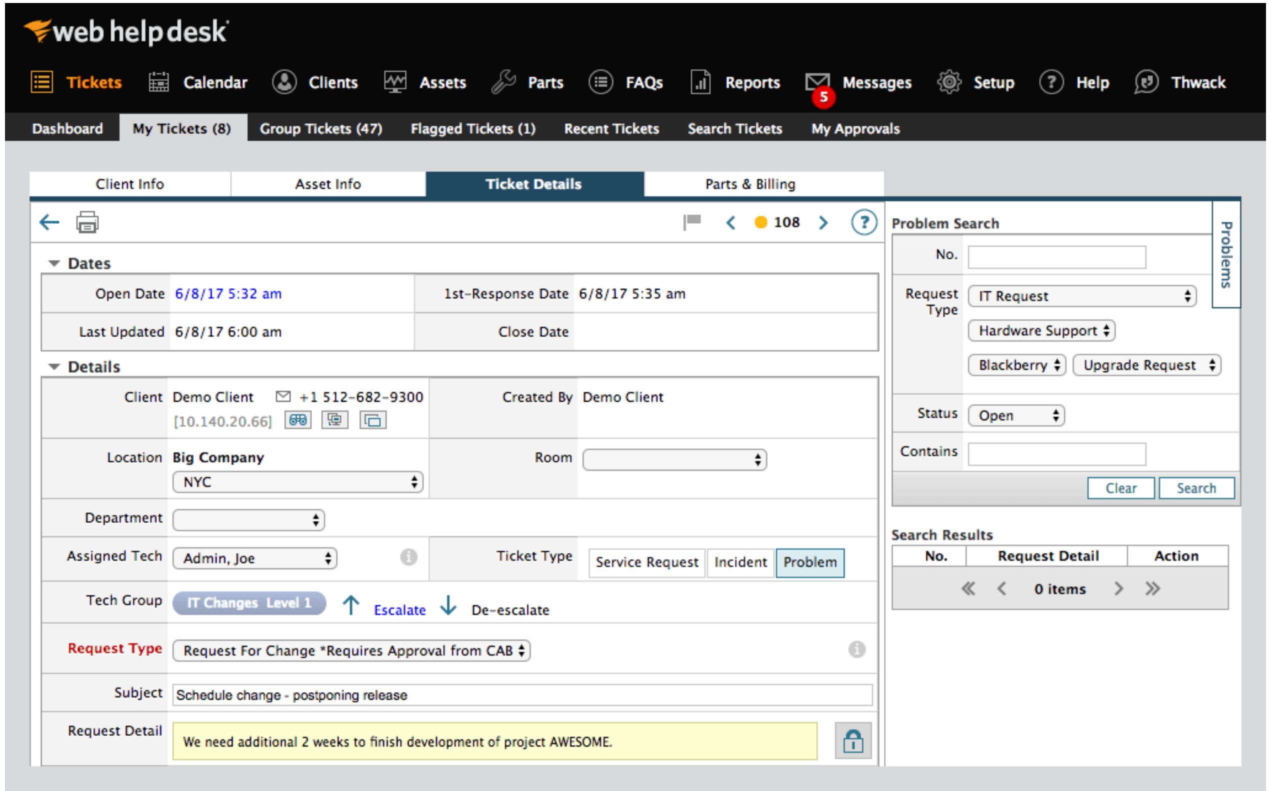
- Common symptoms
  - High network traffic
  - Sluggish response time
  - Data-usage limit notification
  - Limited Internet connectivity
  - No Internet connectivity
  - High number of ads
  - Fake security warnings
  - Unexpected application behavior
  - Leaked personal files/data

4.0

4.1

Documentation and Support Systems

- Ticketing systems



Solarwinds Ticketing System Software

Documentation and Support Systems

- Ticketing systems
  - User information
  - Device information
  - Description of problems
  - Categories
  - Severity
  - Escalation levels
  - Clear, concise written communication
    - Problem description
    - Progress notes
    - Problem resolution

Asset management

- Asset management
  - Inventory lists
  - Database system
  - Asset tags and IDs
  - Procurement life cycle
  - Warranty and licensing
  - Assigned users

Asset management

- Asset management
  - **Barcodes** can make it easier to track items, like how retailers maintain their inventory
  - **Asset Tags** can be used to track misplaced devices by using **RFID (radio frequency identification)**

Barcode sticker

RFID Asset tag

Types of documents

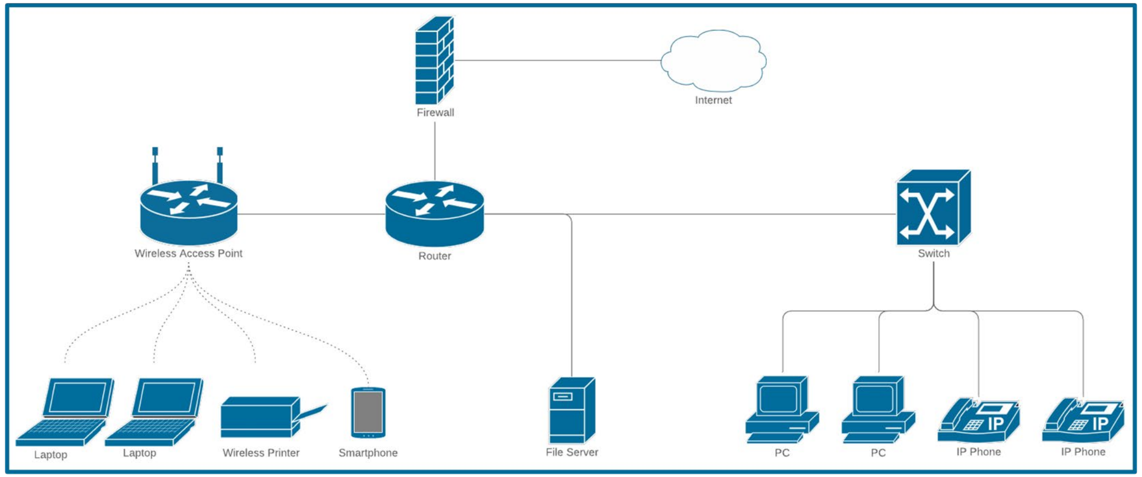
- Acceptable use policies (AUP) define how the employees are allowed to use the services they have access to.
- Regulatory compliance requirements
  - Will influence how an organization has to operate which means IT must also follow the rules of the regulatory bodies
  - Splash screens
- Incident Documentation helps create documentation of what kind of issues are occurring and how they were handled.

Types of documents

- Standard operating procedures (SOP) provide a series of steps to accomplish a certain task
  - Procedures for custom installation of software package
- New-user setup checklist
- End-user termination checklist

Network topology diagram

- Diagrams are used to build the network and it becomes a reference for troubleshoot network issues.



4.2

Change Management

- Documented business processes
  - Rollback plan
  - Sandbox testing
  - Responsible staff member

Change Management

- Change management
  - Request forms
  - Purpose of the change
  - Scope of the change
  - Date and time of the change
  - Affected systems/impact
  - Risk analysis
    - Risk level
  - Change board approvals
  - End-user acceptance

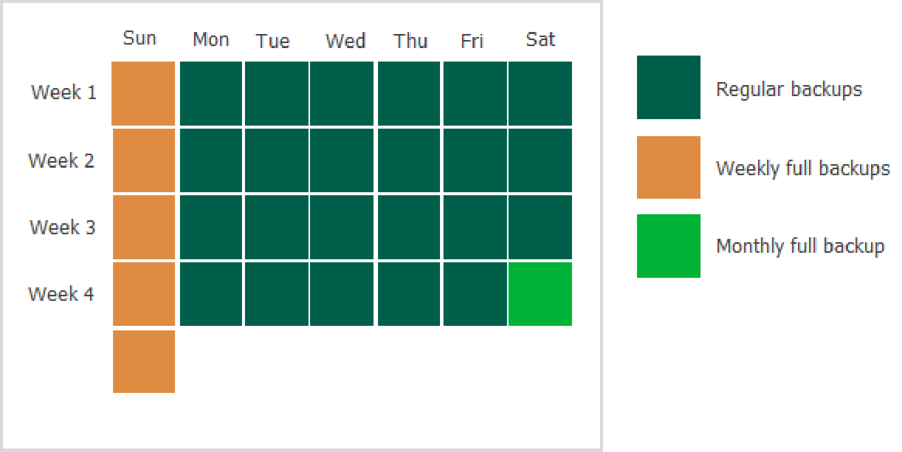
4.3

Backups and Recovery

- Creating a **backup** is the standard operating procedure when working with valuable data.
- Archive bit
  - Show if the file has change
- Backup testing
- Frequency

Backups and Recovery

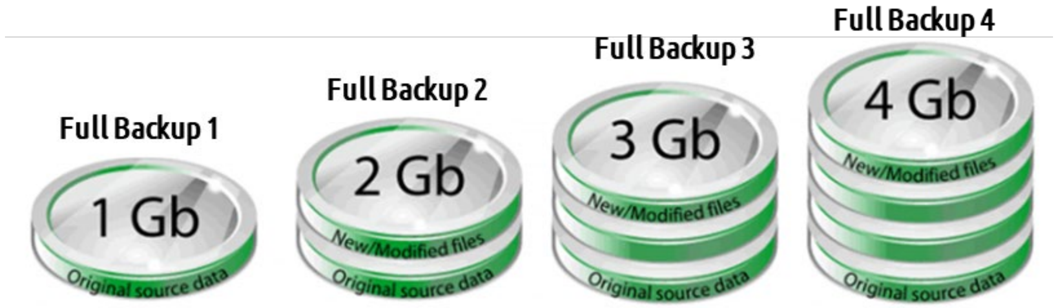
- Backup rotation schemes
  - On site vs. off site
  - 3-2-1 backup rule
    - There should be 3 copies of data; On 2 different media; With 1 copy being off-site.
  - Grandfather-father-son (GFS)



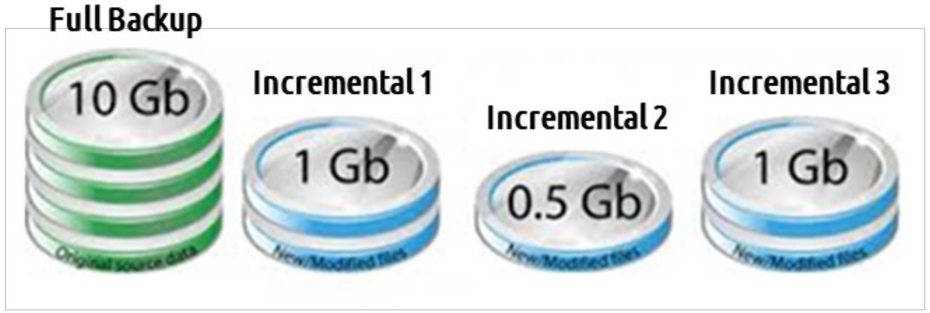
Backups and Recovery

- Full
  - All files are backup
- Incremental
  - Only files that has change since the last backup, clears the archive bit
- Differential
  - Only files that has change since the last full backup, doesn't clear the archive bit
- Synthetic
  - A combo backup that takes a full backup and incremental or differential backup to make another full backup.

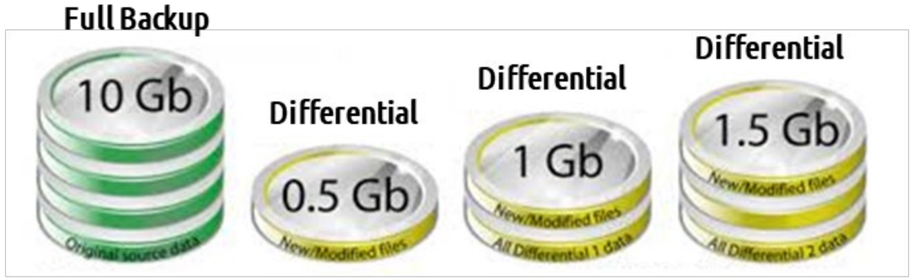
Backups and Recovery



Full Backup contains all new and old files



Note: Each incremental contains only the new/modified files since the last backup



Note: Each differential backup includes all new/modified files since the last backup

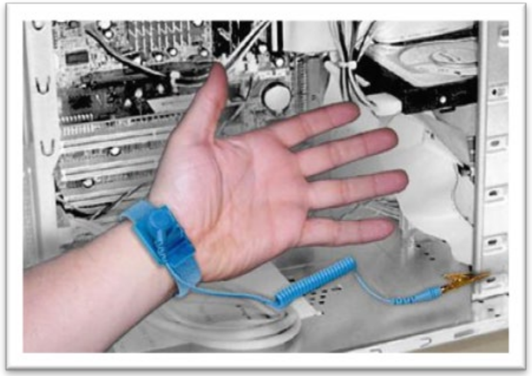
4.4

Safety Procedures

- **Electrostatic discharge (ESD)** is when you discharge the static electricity that naturally builds up in body onto and electronic component. This can be very harmful to the component and can damage it beyond repair.

Safety Procedures

- **Anti-Static Protection Methods**



Anti-static Wrist/Ankle Strap

Anti-static Table Mat

Safety Procedures

- Anti-Static Protection Methods

Anti-static Floor Mat

Safety Procedures

- Anti-Static Protection Methods

Metallized film antistatic bag

Safety Procedures

- Personal safety
- Disconnect power before repairing PC
- Lifting techniques
- Electrical fire safety
- Safety goggles
- Air filtration mask

4.5

Environmental Controls

- Material safety data sheet (MSDS)/documentation for handling and disposal
  - Proper battery disposal
  - Proper toner disposal
  - Proper disposal of other devices and assets

Environmental Controls

SIGMA-ALDRICH sigma-aldrich.com

Material Safety Data Sheet
Version 4.1
Revision Date 10/23/2010
Print Date 02/08/2011

1. PRODUCT AND COMPANY IDENTIFICATION

Product name	:	Chromium(III) acetate hydroxide
Product Number	:	318108
Brand	:	Aldrich
Product Use	:	For laboratory research purposes.
Supplier	:	Sigma-Aldrich Canada, Ltd 2149 Winston Park Drive OAKVILLE ON L6H 6J8 CANADA
Manufacturer	:	Sigma-Aldrich Corporation 3050 Spruce St. St. Louis, Missouri 63103 USA
Telephone	:	+19058299500
Fax	:	+19058299292
Emergency Phone # (For both supplier and manufacturer)	:	1-800-424-9300
Preparation Information	:	Sigma-Aldrich Corporation Product Safety - Americas Region 1-800-521-8956

2. HAZARDS IDENTIFICATION

Emergency Overview

WHMIS Classification
Not WHMIS controlled. Not WHMIS controlled.

GHS Classification
Acute toxicity, Inhalation (Category 4)
Acute toxicity, Dermal (Category 4)
Acute toxicity, Oral (Category 4)
Skin irritation (Category 2)
Eye irritation (Category 2A)
Specific target organ toxicity - single exposure (Category 3)

GHS Label elements, including precautionary statements

Pictogram

Signal word Warning

Hazard statement(s)

H302 + H312	Harmful if swallowed or in contact with skin.
H315	Causes skin irritation.
H319	Causes serious eye irritation.
H332	Harmful if inhaled.
H335	May cause respiratory irritation.

Precautionary statement(s)

P261	Avoid breathing dust/ fume/ gas/ mist/ vapours/ spray.
P264	Wash skin thoroughly after handling.
P270	Do not eat, drink or smoke when using this product.
P271	Use only outdoors or in a well-ventilated area.
P280	Wear protective gloves/ eye protection/ face protection.
P301 + P312	IF SWALLOWED: Call a POISON CENTER or doctor/ physician if you feel unwell.

Aldrich - 318108 Page 1 of 6

Environmental Controls

- Temperature, humidity-level awareness, and proper ventilation
  - Location/equipment placement
  - Dust cleanup
  - Compressed air/vacuums

Environmental Controls

- Power surges, under-voltage events, and power failures
  - Battery backup
  - Surge suppressor

4.6

Incident response

- Chain of custody
  - Is the chronological documentation or paper trail that records the sequence of custody, control, transfer, analysis, and disposition of materials, including physical or electronic evidence.
- Inform management/law enforcement as necessary
- Copy of drive (data integrity and preservation)
  - Bit by bit copy
- Documentation of incident
  - Information about the incident

Licensing

- Licensing/digital rights management (DRM)/end-user license agreement (EULA)
  - Valid licenses
  - Non-expired licenses
  - Personal use license vs. corporate use license
  - Open-source license

Regulations

- Regulated data
  - Credit card transactions
  - Payment Card Industry Data Security Standard is an information security standard for organizations that handle branded credit cards from the major card schemes.
  - Personal government-issued information
- PII (personally Identifiable Information)
  - Personal information about you
  - Address, credit number, social security numbers
  - General Data Protection Regulation (GDPR) is a regulation in EU law on data protection and privacy in the European Union and the European Economic Area

Regulations

- Healthcare data
    - Medicine you take and/or disease
    - Health Insurance Portability and Accountability Act (HIPAA)
    - healthcare information, stipulates how personally identifiable information maintained by the healthcare and healthcare insurance industries should be protected
- Data retention requirements
    - How long to hold the data, usually define by law

4.7

Professionalism

- **Professional appearance and attire**
  - Match the required attire of the given environment
    - Formal
    - Business casual
- **Use proper language** and avoid jargon, acronyms, and slang, when applicable
- **Maintain a positive attitude**/project confidence
- **Actively listen**, take notes, and avoid interrupting the customer

Professionalism

- **Be culturally sensitive**
  - Use appropriate professional titles, when applicable
- **Be on time** (if late, contact the customer)
- **Avoid distractions**
  - Personal calls
  - Texting/social media sites
  - Personal interruptions

Difficult Customers

- **Dealing with difficult customers or situations**
  - Do not argue with customers or be defensive
  - Avoid dismissing customer problems
  - Avoid being judgmental
  - Clarify customer statements (ask open-ended questions to narrow the scope of the problem, restate the issue, or question to verify understanding)
  - Do not disclose experience via social media outlets

Difficult Customers

- **Set and meet expectations/timeline and communicate status with the customer**
  - Offer repair/replacement options, as needed
  - Provide proper documentation on the services provided
  - Follow up with customer/user at a later date to verify satisfaction
- **Deal appropriately with customers' confidential and private materials**
  - Located on a computer, desktop, printer, etc.

4.8

Cases for Scripting

- **Basic automation**
- **Restarting machines**
- **Remapping network drives**
- **Installation of applications**
- **Automated backups**
- **Gathering of information/data**
- **Initiating updates**
- **Other considerations when using scripts**
  - Unintentionally introducing malware
  - Inadvertently changing system settings
  - Browser or system crashes due to mishandling of resources

Script Types and Languages

Language	File Extension	Description
Batch File	.bat	Windows commands saved to a file
PowerShell	.ps1	Replacement for batch files
Visual Basic Script	.vbs	Windows application scripting.
UNIX shell script	.sh	UNIX/Linux/macOS script
Python	.py	Multiplatform flexible language
JavaScript	.js	Commonly used for web programming

4.9

Remote Access Technologies

- **VPN(Virtual Private Network) creates a secure tunnel to a private network over the internet**
  - Required to access resources in a LAN over the Internet
  - Various protocols can be used; PPTP, L2TP, IPSec, OpenVPN, SSL-VPN
- **RDP(Remote Desktop Protocol) used to connect to a Windows desktop over the network**
  - A VPN connection should always be used
  - Never expose port 3389 to the public Internet

Remote Access Technologies

- **Virtual network computer (VNC) is like RDP but multiplatform**
  - Local screen will still be visible
  - Useful to demonstrate something to a user
  - Can potentially reveal confidential data
  - Available on Windows, Linux, or macOS
- **Secure Shell (SSH) provides a secure command line to a remote system**
- **Remote monitoring and management (RMM).**
  - Locally installed agents that can be accessed by a management service provider
- **Microsoft Remote Assistance (MSRA).**
  - A feature that allows a user to view or control a remote Windows computer over a network or the Internet to resolve issues without directly touching the unit

Remote Access Technologies

- **Third-party tools**
  - Screen-sharing software
  - Video-conferencing software
  - File transfer software
  - Desktop management software
- **Security considerations of each access method**